



ICT CONTENT DISTRIBUTION FOR S4

ALL PATHWAY

ACADEMIC YEAR 2025 - 2026

Number of Periods per Week: 3 Periods (33 Weeks)

All Periods per Year: 99

UNIT 1: Google docs. (periods: 6)			
Key Unit Competence: Learners should be able format and collaborate on documents using Google Docs for academic data to effectively			
Week	Contents	Periods/ proposed	Suggested resource for reference
Week 1	Lesson 1: Introduction to Google Docs • Meaning and importance of Google Docs • Accessing Google Docs (via Gmail / Drive) • Navigating the Google Docs interface (menu, toolbar, workspace) • Creating a new document	1	ICT S4 SB P. 145-150
Week 1	Lesson 2: Text Formatting Basics • Changing font style, size, and color • Bold, italics, underline, and highlighting • Alignments (left, center, right, justify) • Line spacing and paragraph formatting Compatibility issues	1	ICT S4 SB P. 150-155
Week 1	Lesson 3: Working with Lists and Tables • Bulleted and numbered lists • Creating and formatting tables • Adjusting rows and columns	1	ICT S4 SB P. 150-155
Week 2	Lesson 4: Inserting Elements • Adding images, drawings, and shapes • Inserting links and bookmarks • Adding headers, footers, and page numbers	1	https://docs.google.com/document/d/1MRcv44ol1BIAEtISV70kdanvw1zEZ494/edit?usp=drive_link&oid=113853035189273884318&rtpof=true&sd=true
Week 2	Lesson 5: Sharing and Collaboration • Sharing settings (view, comment, edit) • Working on the same document with others in real time • Using comments and suggestions	1	https://edu.gcfglobal.org/en/googledocuments/sharing-and-collaborating/1/
Week 2	End Unit Assessment	1	https://bit.ly/3JGmu8K

UNIT 2: Computer hardware (periods: 5)			
Key Unit Competence: Learners should be able to describe the main components of a computer hardware system and solve different hardware issues			
Week	Contents	Periods/ proposed	Suggested resource for reference
Week 3	Lesson 1: External hardware • Input/output devices (keyboard, mouse, monitor, printer) • Describe their functions	1	Computer Science S4 SB, Pg. 18-26
Week 3	Lesson 2: Internal hardware • Internal components: CPU, RAM, motherboard, PSU, etc. • Functions of Internal components	1	ICT S4 SB, Pg. 1-7
Week 3	Lesson 3: Storage devices • Identify storage devices (HDD, SSD, flash drive)	1	Computer Science S4 SB, Pg. 36-40
Week 4	Lesson 4: Computer hardware issues • Hardware common issues (no display, device not detected, overheating, Beeping) • Basic troubleshooting steps • Safety rules when handling hardware	1	ICT S4 SB, Pg. 12-21
Week 4	End Unit Assessment	1	https://bit.ly/3JGmu8K

UNIT 3: Computer software (periods: 4)			
Key Unit Competence: The learner should be able to install, use and solve different software issues.			
Week	Contents	Periods/ proposed	Suggested resource for reference
Week 4	Lesson 1: Computer software types • Define computer software and explain its role. • Identify the main types of computer software.	1	ICT S4 SB for Associate Nursing Program Pg 22-26

	• Give examples of each type of software and explain their uses.		
Week 5	Lesson 2: Software installation and basic use • Operating System installation. • Installation of application program. • Basic features of installed software. • Safety and best practices during installation.	1	ICT S4 SB, Pg. 28
Week 5	Lesson 3: Common software issues • Installation errors • Software freezing • Compatibility issues	1	ICT S4 SB, Pg. 21-27
Week 5	End Unit Assessment	1	https://bit.ly/3JGmu8K

UNIT 4: Programming Algorithm (periods: 18)

Key Unit Competence: Learners should be able to analyze problems and develop algorithms using pseudocode and flowcharts to design logical solutions.

Week	Contents	Periods/ proposed	Suggested resource for reference
Week 6	Lesson 1: Introduction to algorithm • Definition of algorithms • Characteristics of algorithm. • Importance of algorithms in problem-solving	2	Computer Science S4 SB Pg. 132-133
Week 6&7	Lesson 2: Variables and operations • Definition of Variable • Data type • Declaration	2	Computer Science S4 SB Pg. 139-145

	Operators (Arithmetic operators, logical operators, relation operators, increment and decrement operators)		
Week 7	Lesson 3: Input/output - Initialization (variable and constant) - Read and write (statement)	2	Computer Science S4 SB Pg. 145-146
Week 8	Lesson 4: Algorithm representation - Concept of flowchart - Flowchart symbols - Create basic flowcharts from given algorithms or pseudocode. - Convert a written pseudocode into a flowchart format.	2	Computer Science S4 SB Pg. 134-138
WEEK 8 & 9	Lesson 5: Conditional statement - if - if else	2	Computer Science S4 SB Pg. 148-155
Week 9	Nested if	2	Computer Science S4 SB Pg. 148-155
Week 10	Lesson 6: Loop statement for loop	2	Computer Science S4 SB Pg. 157-160
Week 10&11	while loop	2	Computer Science S4 SB Pg. 157-160
Week11	End Unit Assessment	2	https://bit.ly/3JGmu8K

UNIT 5: Python programming (periods: 15)			
Key Unit Competence: Learners should be able to write and execute Python programs that involve decision making and operations on data.			
Week	Contents	Periods/ proposed	Suggested resource for reference
Week 12	Lesson 1: Introduction to Python - Python overview. - Python Installation (IDLE or online editor).	1	Online resource: https://www.tutorialspoint.com/python/python_overview.htm https://www.w3schools.com/python/python_intro.asp

Week 12	Lesson 2: Variables operation • Data types (int, float, str, bool). • Variable (constant and variable declaration) • Operators (Arithmetic operators, Logical operators, relation operators, Increment and Decrement operators)	2	Online resource: https://www.tutorialspoint.com/python/python_data_types.htm https://www.tutorialspoint.com/python/python_variables.htm https://www.w3schools.com/python/python_operators.asp
Week 13	Lesson 3: Input and output functions • Input function • Output function	2	Online resource: https://www.tutorialspoint.com/python/python_user_input.htm

Term 2

Week 1	Lesson 4: Conditional statements • if • if...else • if...elif...else	2	Online resource: https://www.tutorialspoint.com/python/python_if_statement.htm https://www.tutorialspoint.com/python/python_if_else.htm https://www.datacamp.com/tutorial/python-if-elif-else
Week 1	Lesson 5: Nested conditions • nested if	2	Online resources: https://www.tutorialspoint.com/python/nested_if_statements_in_python.htm
Week 2	Lesson 6: While loops statements • while loop	2	Online resource: https://www.tutorialspoint.com/python/python_while_loops.htm

Week 2&3	Lesson 7: For loops statements • for loop	2	Online resource: https://www.tutorialspoint.com/python/python_for_loops.htm
Week 3	End Unit Assessment	2	

UNIT 6: HTML -BASED Web development (periods: 18)			
Key Unit Competence: Learners should be able to design and develop a well-structured web page.			
Week	Contents	Periods/ proposed	Suggested resource for reference
Week 4	Lesson 1: Introduction to HTML • Definition of HTML • Role of HTML • HTML page structure (head and body,)	2	1) Computer Science S4 SB Pg. 303-306
Week 4&5	Lesson 2: Page structure tags • <html>, <head>, <body> • <title>, <meta>, <link>, <!DOCTYPE>	2	Computer Science S4 SB Pg. 307-309
Week 5&6	Lesson 3: Text formatting tags Control how text appears: • Headings: <h1> to <h6> • Paragraphs: <p> • Emphasis: , <i>, <u>, • marque	2	1) Computer Science S4 SB Pg. 310-311
Week 7	Lesson 4: List tags • Unordered List , • Ordered List , • List	2	Computer Science S4 SB Pg. 317-320

Week 7 & 8	Lesson 5: Media tags • Embed images , • Audio <audio>, • Video: <video>,	2	1) Computer Science S4 Pg. 322-323 2) Online resource: https://www.tutorialspoint.com/html/html_audio_tag.htm https://www.tutorialspoint.com/html/html_video_tag.htm
Week 8	Lesson 6: Table tags Structure tabular data: • <table> • <tr> • <td> • <th>.	2	Computer Science S4 Pg. 326-328
Week 9	Lesson 7: Form elements tags Handle user input: • <form> • <input> • <text area> • <button> • <select> • <option>	2	Computer Science S4 Pg. 328-333
Week 9 & 10	Lesson 8: Link and navigation tags Enable hyperlinks and navigation: • <a>	2	Computer Science S4 Pg. 323-326
Week 10	End Unit Assessment	2	

UNIT 7: Cascading style sheet (periods: 8)

Key Unit Competence: Learners should be able to style and enhance the appearance of web pages.

Week	Contents	Periods/ proposed	Suggested resource for reference
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Week 10	Lesson 1. Text formatting • Font styling • Text appearance • Spacing & layout • Effects	2	Computer Science S4 Pg. 345-347, 350-351, 354-356
Week 11	Lesson 2: Layout styling • Borders and padding elements (width, height, margin, padding, border, box-sizing:) • Background of elements (background-color, background-image, background-repeat, background-size, background-position)	2	Computer Science S4 Pg.337-338, 359-360, 364-366

TERM 3

Week 1	Lesson 3: Linking CSS to HTML • Internal CSS • External CSS	2	Computer Science S4 Pg.351-353
Week 1&2	End Unit Assessment	2	

UNIT 8: Introduction to AI (periods: 9)			
Key Unit Competence: Learners should be able to explain the basic concepts, types and applications of Artificial Intelligence (AI) while reflecting on its ethical implications.			
Week	Contents	Periods/ proposed	Suggested resource for reference
Week 2	Lesson 1 Basics concepts of AI • Definition of artificial intelligence (AI) • Identify key characteristics of AI systems • Comparison of human and machine intelligence • Importance of AI • Types of AI (Narrow AI, general AI, super AI).	2	Online resources: https://www.tutorialspoint.com/artificial_intelligence/index.htm

			https://www.tutorialspoint.com/artificial_intelligence/artificial_intelligence_types.htm
Week 3	Lesson 2: Applications of AI • Education • Healthcare • Agriculture • Transport • Industry	2	Online resource: https://www.tutorialspoint.com/artificial_intelligence/artificial_intelligence_applications.htm
Week 3&4	Lesson 3: AI ethics • Definition of ethics in Technology • Humans 'Role in AI • Ethical and Social Concerns (Bias in AI, Privacy and Surveillance,)	2	Online resources: https://ethics-of-ai.mooc.fi/start https://www.tutorialspoint.com/artificial_intelligence/artificial_intelligence_ethics_and_bias.htm
Week 4	Lesson 2: AI tools for solving problems • Definition of AI tools • Categories and role of AI (Vision AI, Text AI, Voice AI, Robotics) • Example of problem solved by AI (school attendance, language translation, face recognition) • Use AI Tools: Text AI, Voice AI.	2	https://chatgpt.com/ https://notegpt.io/ai-voice-translator
Week 5	End Unit Assessment	1	

UNIT 9: Database design (periods: 16)			
Key Unit Competence: Learners should be able to explain database concepts and create a relational model			
Week	Contents	Periods/ proposed	Suggested resource for reference
Week 5	Lesson 1. Database concepts: • Definition, • Database access levels and users • Physical independence • Relational database model • Primary key • Foreign key	2	Computer Science S5 Pg. 126-130
Week 6	Lesson 2. Types of relationship (1-to-1, 1-to-Many, Many-to Many)	2	Computer Science S5 Pg. 165-167
Week 6&7	Lesson 3. Entity relationship diagram • Entities • Attributes • Relationship • Representation of (E-R-D) Diagram	2	Computer Science S5 Pg. 156-163
Week 7	Lesson 4: Relational model Tables, columns, rows, primary keys, foreign keys, Data types	2	Computer Science S5 Pg. 144-147
Week 7	Lesson 5: Keys in relational model • Tables creation (with right data types and constraints: Primary key, foreign key) • Relation model representation (ERD to Relation model)	2	Computer Science S5 Pg. 148-162
Week 8	Lesson 6: Normalization • Normalization definition • Normalization role • Types of Normal Forms	1	Computer Science S5 Pg. 171-180

Week 8	Lesson 7: First and Second Normal Forms	2	Computer Science S5 Pg. 172-175
Week 9	Lesson 8: Third Normal Form	2	Computer Science S5 Pg. 175-176
Week 9	End Unit Assessment	1	

ICT S4: END OF UNITS ASSESSMENT SAMPLE

UNIT 1: Google Docs (/20 marks)

Knowledge and understanding

2 marks each

1. Define Google Docs.
2. Give two advantages of using Google Docs
3. Which menu is used to insert a table?
 - a) Tools
 - b) Insert
 - c) Format
 - d) File
4. Google Docs automatically saves your work (True/False)
5. Differentiate between “view” and “edit” permissions in Google docs.

Skills

1 marks each

6. Create and rename a new document and share it to your colleagues.
7. Apply bold, italics, and underline to text in that google document.
8. Insert a picture from your computer.
9. Insert a table with 3 rows and 2 columns.

Attitudes & Values

3 marks each

10. Why is it important to respect others’ edits?
11. It is ethical to delete peers’ text in a shared doc. (True/False)